DPT INFORMATIQUE
COMPUTER SCIENCE AND INFORMATION TECHNOLOGY
INSA Campus LyonTech - 7 Avenue Jean Capelle
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Knowledge Engineerin

Semantic Web

IDENTIFICATION

CODE :	IF-4-WS
ECTS :	2.0

HOURS

Lectures :	9.0 h
Seminars :	4.0 h
Laboratory :	12.0 h
Project :	0.0 h
Teacher-student contact :	25.0 h
Personal work :	10.0 h
Total :	35.0 h

ASSESSMENT METHOD

A 1.5h exam will test your knowledge (documents allowed). Your mini-project will be evaluated by groups of six students: you will provide an experiment report and will defend your project to an audience (teachers and other students). The final mark is computed as follow: 60% for the exam ; 40% for the project.

TEACHING AIDS

All documents are available on MOODLE (http://moodle.insa-lyon.fr).

TEACHING LANGUAGE

French

CONTACT

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AIMS

How to formalize and exchange information and knowledge on the Web? To answer this question, this class introduces Semantic Web which can be understood as an extension of the classic Web by allowing to exchange et reuse data far beyond the boundaries of a Web site or an application thanks to a "universal language". For that, the World Wide Web Consortium (W3C) introduced a series of standards for representing, interrogating, exchanging and reasoning on knowledge (data format, languages, protocoles and description logics). Although the original and complete vision of Tim Berners-Lee (founder of the W3C) may remain utopian for some practitioners today, a big step has already been achieved through the Web of Data, structuring and linking existing information on the Web (Linked data). Applications are numerous and close to open data initiatives that flourish in many companies and territorial collectivities.

In this context, this class aims at:
- Learn the main W3C standards on semantic Web technologies
- Be able to represent knowledge with these standards
- Be able to interrogate Web data
- Be able to use reasoning mechanisms on Web data
- Be able to develop a system using these notions along with Web services (API) to give semantics to existing Web pages and propose a use case

CONTENT

In this class, you will study the theoretical foundations of Semantic Web to represent basic information (RDF) and query knowledge bases (SPARQL). You will also see how to represent information with richer langages (RDF-S et OWL) and a few reasoning mechanisms (RIF). This will be done through 5 sessions of 1.5 hours each. During the unique exercice session (4 hours) you will study in detail how to represent both formally and concretely (XML, JSON) information and statements of the real world with RDF graphs. You will also learn how to query such representations with the graph matching technique and will test in practice with a DBpedia access point (where formally lie Wikipedia information). During 3 practical sessions, you will prepare a mini-project with the following aim: adding semantics to the results of a search engine (Google, Bing, ...). For that, you will learn how to interrogate the search engine and inspect the HTML content: each WEB page of the results will turned into a RDF graph. You will then be able to compare the Web pages and, for example, group pages with a similar content with respect to their semantics and not their syntax as it is usually done in the classical Web.

BIBLIOGRAPHY

- GANDON Fabien et al. Le web sémantique : comment lier les données et les schémas sur le Web. Paris, Dunod, 2012.
- ALLEMANG Dean et HENDLER James. Semantic Web for the Working Ontologist. Effective Modeling in RDFS and OWL. Morgan Kaufmann, 2011.
- MOOC Web sémantique et Web de données de Gandon, Corby, Faron Zucker https://www.france-universite-numerique-mooc.fr/

PRE-REQUISITE

This class does not require strong prior knowledge. A plus is to have studied XML (IF-3-BDS). It is strongly advise however to follow the class on artificial intelligence and logics (IF-4-ALIA) during the semester.

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